

Machine Learning

UNIT – I: Introduction to Machine Learning

1. Define Machine Learning. Explain its types (Supervised, Unsupervised, Reinforcement Learning).
2. What are the key differences between AI, Machine Learning, and Deep Learning?
3. Explain the steps involved in a typical Machine Learning workflow.
4. What is the difference between classification and regression?
5. Define the following terms:
 - a) Model
 - b) Training data
 - c) Testing data
 - d) Feature
 - e) Label

UNIT – II: Supervised Learning

1. What is Supervised Learning? List its applications.
2. Explain Linear Regression with an example and code logic.
3. What is Logistic Regression? When is it used?
4. Explain the K-Nearest Neighbors (KNN) algorithm.
5. Compare Linear Regression and Logistic Regression.

UNIT – III: Unsupervised Learning

1. What is Unsupervised Learning? Explain its real-world applications.
2. Describe the K-Means Clustering algorithm with steps.
3. What is Hierarchical Clustering? Explain Agglomerative method.
4. What is Dimensionality Reduction? Explain PCA.
5. Compare K-Means and DBSCAN clustering

UNIT – IV: Evaluation Metrics & Model Selection

1. What is Overfitting and Underfitting? How to prevent them?
2. Explain confusion matrix and derive Accuracy, Precision, Recall, F1-score.
3. What is Cross-validation? Why is it important?
4. Discuss the concept of Bias-Variance tradeoff.
5. Explain ROC curve and AUC.

UNIT – V: Machine Learning in Python

1. List important Python libraries for Machine Learning (NumPy, Pandas, Matplotlib, Scikit-learn).
2. Write Python code to load dataset, train a model, and evaluate accuracy.
3. How to implement Linear Regression using scikit-learn?
4. Write Python code for K-Means Clustering and plot the clusters.
5. Explain the role of preprocessing techniques like scaling and encoding.